WoSM Academy

MAT231-F27 · CALCULUS II

Stirling's approximation

How large is $n!$?

Dr. Ali Khudiyev
WoSM Academy

Lecture 08 · Unit 8 · Fall 2027

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Last time

- A generating function $A(x) = \sum a_n x^n$ holds a whole sequence.
- Products count combined choices; linear recurrences become rational functions.
- Catalan numbers: $C_n = \frac{1}{n+1}\binom{2n}{n}$, growing like 4^n.

Two promises are still open: why $\binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}}$, and where the factor $n^{-3/2}$ in C_n comes from. Both follow from one formula for $n!$.

This lecture

01 How large is $n!$?

02 $\ln n!$ as a sum

03 The constant $\sqrt{2\pi}$

04 Using Stirling's formula

05 Summary and practice

Reading: TODO(owner): choose a text for this unit — the books on the module's reading list do not cover Stirling's formula.

01

How large is $n!$?

Factorials grow fast

n	$n!$	
10	$3\,628\,800$	
20	$\approx 2.43 \cdot 10^{18}$	
52	$\approx 8.07 \cdot 10^{67}$	(orderings of a deck of cards)
100	$\approx 9.33 \cdot 10^{157}$	

- $n!$ counts the orderings of n objects: trying them all is hopeless beyond $n \approx 15$.
- A comparison sort must tell apart all $n!$ orderings, so it needs at least $\log_2 n!$ comparisons in the worst case (Section 4).

We want a simple formula for $n!$, or at least for $\ln n!$, that is accurate for large n.

First, crude bounds

- **Above:** $n! = 1 \cdot 2 \cdots n \le n \cdot n \cdots n = n^n$.
- **Below:** $e^n = \sum_{k \ge 0} \frac{n^k}{k!} \ge \frac{n^n}{n!}$, so $n! \ge \left(\frac{n}{e}\right)^n$.

Taking logarithms:

$$n \ln n - n \;\le\; \ln n! \;\le\; n \ln n,$$

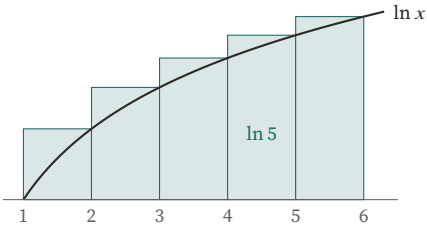
so $\ln n! = n \ln n + O(n)$, and in O-notation $\log n! = \Theta(n \log n)$.

> **Remark**
>
> That is enough for many uses in computer science. But $n!$ itself is still only known up to a factor of e^n. Today: $n!$ up to a factor that tends to 1.

02

$\ln n!$ as a sum

Sum against integral



$\ln n! = \sum_{k=1}^{n} \ln k$, and ln is increasing.

On $[k-1, k]$, $\ln x \le \ln k$: the rectangles lie above the graph. On $[k, k+1]$, $\ln x \ge \ln k$. So (as in Lecture 2 and Calculus I, Unit 26)

$$\int_1^n \ln x \, \mathrm{d}x \le \ln n! \le \int_1^{n+1} \ln x \, \mathrm{d}x.$$

With $\int \ln x \, \mathrm{d}x = x \ln x - x$:

$$n \ln n - n + 1 \;\le\; \ln n! \;\le\; (n+1)\ln(n+1) - n.$$

The two sides differ by about $\ln n$: so $\ln n! = n \ln n - n + O(\ln n)$.

Trapezoids do better

The trapezoidal rule (Calculus I, Unit 25) on $[k, k+1]$ uses the chord from $(k, \ln k)$ to $(k+1, \ln(k+1))$:

$$\int_k^{k+1} \ln x \, dx = \frac{\ln k + \ln(k+1)}{2} + E_k.$$

- ln is concave, so the chord lies below the graph: $E_k > 0$.
- The error bound for the trapezoidal rule, with $|(\ln x)''| = \frac{1}{x^2} \le \frac{1}{k^2}$: $E_k \le \frac{1}{12k^2}$.

Add up for $k = 1, \ldots, n-1$. The chords give $\ln n! - \frac{1}{2}\ln n$, so

$$n \ln n - n + 1 = \ln n! - \tfrac{1}{2}\ln n + \sum_{k=1}^{n-1} E_k.$$

A constant appears

Rearranged:

$$\ln n! - \left(n + \frac{1}{2}\right)\ln n + n = 1 - \sum_{k=1}^{n-1} E_k.$$

- $0 < E_k \le \frac{1}{12k^2}$, so $\sum E_k$ converges (comparison test, Lecture 2). Call its sum E.
- So the left-hand side tends to $c = 1 - E$, and

$$n! \;\sim\; e^c \sqrt{n}\left(\frac{n}{e}\right)^n,$$

where $a_n \sim b_n$ means $\frac{a_n}{b_n} \to 1$.

A convergent series gave us the shape of the answer. What is the constant e^c?

03

The constant $\sqrt{2\pi}$

Wallis's integrals

Let $I_n = \int_0^{\pi/2} \sin^n x \, \mathrm{d}x$. Integration by parts gives $I_n = \frac{n-1}{n} I_{n-2}$, with $I_0 = \frac{\pi}{2}$, $I_1 = 1$, so

$$I_{2m} = \frac{\pi}{2} \cdot \frac{1 \cdot 3 \cdots (2m-1)}{2 \cdot 4 \cdots (2m)} = \frac{\pi}{2} \cdot \frac{\binom{2m}{m}}{4^m}, \qquad I_{2m+1} = \frac{2 \cdot 4 \cdots (2m)}{3 \cdot 5 \cdots (2m+1)}.$$

As $0 \le \sin x \le 1$, the I_n decrease: $I_{2m+1} \le I_{2m} \le I_{2m-1}$. And $\frac{I_{2m+1}}{I_{2m-1}} = \frac{2m}{2m+1} \to 1$, so $\frac{I_{2m}}{I_{2m+1}} \to 1$. Multiplying out:

$$\frac{\pi}{2}(2m+1)\left(\frac{\binom{2m}{m}}{4^m}\right)^2 \to 1, \qquad \text{so} \qquad \binom{2m}{m} \sim \frac{4^m}{\sqrt{\pi m}}.$$

The first promise of Lecture 7.

The constant

Now compute $\binom{2m}{m} = \frac{(2m)!}{(m!)^2}$ with $n! \sim e^c\sqrt{n}\,(n/e)^n$:

$$\binom{2m}{m} \sim \frac{e^c\sqrt{2m}\,(2m/e)^{2m}}{e^{2c}\,m\,(m/e)^{2m}} = \frac{\sqrt{2}}{e^c}\cdot\frac{4^m}{\sqrt{m}}.$$

Compare with $\frac{4^m}{\sqrt{\pi m}}$: $\frac{\sqrt{2}}{e^c} = \frac{1}{\sqrt{\pi}}$, so $e^c = \sqrt{2\pi}$.

Theorem (Stirling's formula)

$$n! \;\sim\; \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

More precisely, for every $n \geq 1$: $1 < \dfrac{n!}{\sqrt{2\pi n}\,(n/e)^n} < e^{\frac{1}{12}\left(\frac{1}{n}+\frac{1}{n^2}\right)}$.

Where the error bound comes from

From the two previous frames,

$$\ln \frac{n!}{\sqrt{2\pi n}\,(n/e)^n} = \sum_{k \geq n} E_k,$$

the tail of the convergent series $\sum E_k$. With $0 < E_k \leq \frac{1}{12k^2}$ and the integral test bound (Lecture 2):

$$0 < \sum_{k \geq n} E_k \leq \frac{1}{12} \sum_{k \geq n} \frac{1}{k^2} \leq \frac{1}{12}\left(\frac{1}{n^2} + \int_n^{\infty} \frac{\mathrm{d}x}{x^2}\right) = \frac{1}{12}\left(\frac{1}{n} + \frac{1}{n^2}\right).$$

A sharper argument (H. Robbins, 1955) gives the upper bound $e^{1/(12n)}$.

How good is it?

n	$n!$	$\sqrt{2\pi n}\,(n/e)^n$	ratio
1	1	0.922	1.0844
5	120	118.02	1.0168
10	3 628 800	3 598 696	1.0084
20	$2.4329 \cdot 10^{18}$	$2.4228 \cdot 10^{18}$	1.0042
100	$9.3326 \cdot 10^{157}$	$9.3248 \cdot 10^{157}$	1.00083

- The *relative* error is about $\frac{1}{12n}$: below 1% from $n = 10$ on.
- The *absolute* error grows without bound. "~" is a statement about ratios.

04

Using Stirling's formula

Which precision do you need?

For $n = 10$, $\ln 10! = 15.104$:

Approximation of $\ln n!$	Error term	$n = 10$
$n \ln n$	$O(n)$	23.026
$n \ln n - n$	$O(\ln n)$	13.026
$(n + \frac{1}{2}) \ln n - n + \ln \sqrt{2\pi}$	$O(\frac{1}{n})$	15.096

- For growth rates, $\Theta(n \log n)$ is enough.
- For counting, probabilities and constants, use the full formula.

Binomial coefficients and Catalan numbers

- $\binom{2n}{n} \sim \dfrac{4^n}{\sqrt{\pi n}}$, and therefore $C_n = \dfrac{1}{n+1}\binom{2n}{n} \sim \dfrac{4^n}{\sqrt{\pi}\, n^{3/2}}$: the second promise of Lecture 7.

> **Example (Fair coins)**
>
> The probability of exactly n heads in $2n$ tosses of a fair coin is $\binom{2n}{n}/4^n \approx \frac{1}{\sqrt{\pi n}}$. For 100 tosses: $\frac{1}{\sqrt{50\pi}} \approx 0.0798$ (exact: 0.0796). The most likely outcome is unlikely.

The $\sqrt{2\pi}$ here is the same as in the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$ (Unit 21) — and in the normal distribution of Probability Theory and Statistics.

Sorting needs $n \log_2 n$ comparisons

- A comparison sort is a binary decision tree: each comparison has two outcomes. Its leaves must include all $n!$ orderings, and a binary tree of height h has at most 2^h leaves. So the worst case needs $h \geq \log_2 n!$ comparisons.

By Stirling's formula,

$$\log_2 n! = n \log_2 n - n \log_2 e + \tfrac{1}{2} \log_2 (2\pi n) + o(1) \approx n \log_2 n - 1.443\, n.$$

> **Example** ($n = 1000$)
>
> $\log_2 1000! \approx 8529.4$: every comparison sort needs at least 8530 comparisons in the worst case. Merge sort uses at most $n\lceil \log_2 n \rceil = 10\,000$.

Counting digits

The number of decimal digits of $n!$ is $\lfloor \log_{10} n! \rfloor + 1$, and

$$\log_{10} n! \approx \left(n + \frac{1}{2}\right) \log_{10} n - n \log_{10} e + \log_{10} \sqrt{2\pi}.$$

- 52! has 68 digits.
- 1000! has 2568 digits.

No need to compute 1000! itself: Stirling's error factor $e^{1/12000}$ changes $\log_{10} 1000!$ by less than 10^{-4}.

05

Summary and practice

Summary

- Crude: $(n/e)^n \le n! \le n^n$, so $\log n! = \Theta(n \log n)$.
- $\ln n! = \sum \ln k$, compared with $\int \ln x \, \mathrm{d}x$: $\ln n! = n \ln n - n + O(\ln n)$.
- Trapezoids: the errors $E_k \le \frac{1}{12k^2}$ form a convergent series, so $n! \sim e^c \sqrt{n}\,(n/e)^n$ for some constant c.
- Wallis's integrals give $\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$ and $e^c = \sqrt{2\pi}$.
- **Stirling**: $n! \sim \sqrt{2\pi n}\,(n/e)^n$, with relative error about $\frac{1}{12n}$.
- Uses: central binomial and Catalan numbers, coin tosses, the sorting lower bound, sizes of huge numbers.

Practice

1. Estimate $30!$ with Stirling's formula and compare with $30! \approx 2.6525 \cdot 10^{32}$.
2. Show that $\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$ follows from Stirling's formula.
3. Estimate $\binom{100}{50}$.
4. How many decimal digits does $100!$ have?
5. Find $\displaystyle\lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n}$.
6. For large n, which is larger: $n!$ or $(n/2)^n$? $n!$ or $(n/3)^n$?
7. Use $\ln n! = n\ln n - n + O(\ln n)$ to find the exponential growth rate of $\binom{3n}{n}$, that is, $\lim \sqrt[n]{\binom{3n}{n}}$.

Answers

1. $\sqrt{60\pi}\,(30/e)^{30} \approx 2.6452 \cdot 10^{32}$: ratio 1.0028, close to $e^{1/360}$.
2. $\dfrac{(2n)!}{(n!)^2} \sim \dfrac{\sqrt{4\pi n}\,(2n/e)^{2n}}{2\pi n\,(n/e)^{2n}} = \dfrac{4^n}{\sqrt{\pi n}}$.
3. $\dfrac{4^{50}}{\sqrt{50\pi}} \approx 1.011 \cdot 10^{29}$ (exact: $1.0089 \cdot 10^{29}$).
4. 158.
5. $\frac{1}{e}$: $\sqrt[n]{n!} \sim \frac{n}{e}\,(2\pi n)^{1/(2n)}$, and $(2\pi n)^{1/(2n)} \to 1$.
6. $\frac{n!}{(n/2)^n} \sim \sqrt{2\pi n}\left(\frac{2}{e}\right)^n \to 0$, so $(n/2)^n$ is larger; $\frac{n!}{(n/3)^n} \sim \sqrt{2\pi n}\left(\frac{3}{e}\right)^n \to \infty$, so $n!$ is larger.
7. $\ln\binom{3n}{n} = 3n\ln 3n - n\ln n - 2n\ln 2n + O(\ln n) = n\ln\frac{27}{4} + O(\ln n)$, so the limit is $\frac{27}{4}$.

Next time

Lecture 09 — Functions of several variables (Unit 9)

- Part 2 of the module begins: calculus in $\mathbb{R}^n$
- Norms and distance; open and closed sets
- Limits and continuity — and why approaching along lines is not enough

Reading: G. Strang, E. Herman, *Calculus, Volume 3* (OpenStax), Sections 4.1 and 4.2.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.